

Thermodynamic analysis of nanoparticle size effect on kinetics in Fischer–Tropsch synthesis by lanthanum promoted iron catalyst

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Abstract The kinetic parameters of the Fischer–Tropsch synthesis (FTS) on iron catalyst are analyzed by size-dependent thermodynamic method. A Langmuir–Hinshelwood kinetic equation is considered for evaluation of catalytic activity of lanthanum promoted iron catalyst. A series of unsupported iron catalysts with different particle sizes were prepared via microemulsion method. The experimental results showed that catalyst activity pass from a maximum value by increasing the iron particle size. Also, data presented that iron particle size has considerable effects on adsorption parameters and FTS rates. The ratio of surface tension (σ) to nanoparticle radius (r) is important in FTS reaction on iron catalyst. Finally, the results showed that by increasing of iron particle size from 18 to 45 nm the activation energies of catalysts and heats of adsorption of catalysts as two main parameters of FTS reaction increased from 89 to 114 kJ/mol and from 51 to 71 kJ/mol, respectively.

Abbreviations

FTS	Fischer Tropsch synthesis
r_{FTS}	Rate of FTS reaction
k	Rate constant of FTS reaction
b	Adsorption parameter

r	Catalyst particle size
P_{H_2}	Partial pressure of hydrogen
$P_{\text{H}_2\text{O}}$	Partial pressure of water
σ	Surface tension of solid catalyst
E_a	Activation energy for the overall catalytic process
ΔH	Adsorption enthalpy
k_∞	Size independent FTS reaction rate constant
η	A parameter which is equal to $\eta = 2\sigma V_M/RT$
b_∞	Size independent adsorption parameter
χ	Brønsted–Polanyi parameter, $0 < \chi < 1$
δ_∞	Absolute temperature independent surface tension energies
ΔH_∞	Size independent adsorption enthalpy
$E_{a\infty}$	Size independent activation energy

1 Introduction

Strong interest in the Fischer–Tropsch synthesis (FTS) which converted synthesis gas into hydrocarbons has attracted extensive attention [1–3]. FTS reaction is a surface phenomenon and for optimum catalyst performance, maximum metal loaded must be used. A rule of thumb in heterogeneous catalysis is that the smaller metal crystallites provide the largest surface area on which the reaction may happen [4–8]. Therefore, it is expected that the smaller metal crystallite size will be more active under test conditions. The FTS reaction may be affected by the catalyst structures and geometry, reaction condition, solubility of products and feed in wax coated of active site and many other parameters [9–12]. Metal crystallite size had great effects on catalyst structures and geometry. Thus, nanoparticles often have superior or even new catalytic properties following from their nanometric size that give them increased surface-to-volume ratios and modified chemical

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potentials, compared to their bulk counterparts [5, 13, 14]. Since the past decade has seen major advances in understanding the structure sensitivity of transition metal-catalyzed surface reactions, the kinetic analysis helps to understand how structure sensitivity affects FTS activity. The FTS kinetics has been extensively studied, and many attempts have been made for the rate equations describing the FTS reactions [15–19].

Murzin [20–22] reported that FTS reaction on supported cobalt-based catalyst shows high sensitivity to cobalt particle size. He developed a useful thermodynamic method for evaluation of structure sensitivity of heterogeneous catalytic reaction. Thermodynamic analysis of the effect of the nanoparticle size on the adsorption equilibrium, chemical kinetics and rates was initiated by Parmon [23] and developed by Murzin [20–22]. The key in thermodynamic analysis is description of influence of the nanometric size of particles on the chemical potential of the active phase (e.g., clusters supported on a carrier). Parmon and Murzin considered that the decrease of the nanocluster size affects the chemical potential of the active phase due to excessive surface energy and increases effects on the internal Laplacian pressure.

New results showed that nano-sized iron particles play an essential role to achieve high FTS activity [11, 14, 24–28]. Iron-based catalysts are preferred for utilizing synthesis gas derived from coal or biomass because of the excellent activity for the water–gas-shift reaction, which allows using a synthesis gas with a low H_2/CO ratio directly without an upstream shift-step [3, 15, 29, 30]. Iron-based catalysts are usually used as massive catalyst in FTS reaction [3, 15, 25, 29, 31–33]. The major inconveniences related to the employment of massive catalysts are their physical degradation and low surface area. In our previous works, we studied the effect of nano particles on physico-chemical, textural properties and catalytic activity of iron catalysts in FTS reaction [13, 24, 34–37].

The objective of this work is to provide a theoretical basis for size dependence evaluation of FTS reaction on lanthanum promoted iron catalyst. For achieving this purpose, a series of unsupported iron catalysts with different particle sizes were prepared via microemulsion methods. A suitable FTS kinetic rate equation is used for describing the rate of CO consumption. The size dependency of kinetic parameters was evaluated using experimental results and Parmon and Murzin method.

2 Experimental

2.1 Catalyst preparation and characterization

The Fe/Cu/La catalysts with different particle size were prepared as described previously [24, 34, 35]. The catalysts

composition was designated in terms of the atomic ratios as: 100Fe/5.64Cu/2La. Characterization of catalysts was reported in previous works [34, 35].

2.2 Experimental apparatus and procedure

Steady-state FTS reaction rates were measured in a continuous spinning basket reactor. A detailed description of the experimental setup and procedures has been provided in our previous works [17]. The fresh catalyst was crushed and sieved to particles with the diameter of 0.25–0.36 mm (40–60 ASTM mesh). The weight of the catalyst loaded was 2.5 g, which diluted by 30 cm³ inert silica sand with the same mesh size range. The catalyst samples were activated by a 5 % (v/v) H_2/N_2 gas mixture with space velocity equal to 15.1 nl h⁻¹ g_{Fe}⁻¹ at 1 bar and 1,800 rpm. The reactor temperature increased to 673 K with a heating rate of 5 K/min, maintained for 1 h at this temperature, and then reduced to 543 K. The activation was followed by the synthesis gas stream with H_2/CO ratio of 1 and space velocity equal to 3.07 nl h⁻¹ g_{Fe}⁻¹ for 24 h at 1 bar and 543 K.

After catalyst activation, synthesis gas was fed to the reactor at operating conditions of 563 K, 17 bar, H_2/CO ratio of 1 and a space velocity equal to 10.4 nl h⁻¹ g_{Fe}⁻¹. After reaching steady state, the FTS reaction rate was measured. Experimental conditions were varied in the following ranges: pressure = 13–25 bar, temperature = 543–603 K, GHSV = 2.8–14 nl h⁻¹ g_{cat}⁻¹, and H_2/CO feed ratio = 0.5–2.0. Each experiment was replicated three times to verify the experimental data accuracy and reproducibility.

The out gas was analyzed by a gas chromatograph (Varian CP-3800) equipped with TCD and FID detectors. The CO, CO₂, N₂, and O₂ were analyzed through two packed column in series (Molecular sieve 13 × CP 81025 with 2 m length, and 3 mm OD, and Haysep Q CP1069 with 4 m length, and 3 mm OD) connected to TCD detector. The C₁–C₅ hydrocarbons were analyzed via a capillary column (CP fused silica with 25 m × 0.25 mm × 0.2 μm film thickness) connected to FID detector. Hydrogen was analyzed through Shimadzu, GC PTF 4C, equipped with TCD detector and two column in series (Propack-Q with 2 m length, and 3 mm OD for CO₂, C₂H₄ and C₂H₆ separation and molecular sieve-5A with 2 m length, and 3 mm OD for CO, N₂, CH₄, and O₂ separation), which were connected to each other via a three-way valve.

The collected liquid (Including hydrocarbons and oxygenates) were analyzed offline with Varian CP-3800 gas chromatograph equipped with capillary column (TM DH fused silica capillary column, PETRO COL 100 m × 0.25 mm × 0.5 μm film thickness) connected to FID detector.

Conversion of carbon monoxide and hydrogen, and formation of various products were measured within a time

period of 24 h at each run. At regular intervals, the standard experiment was repeated to control possible deactivation of the catalysts. For each operation condition, it took at least 12 h to ensure the steady-state behavior of the catalyst after a change in the reaction condition. Total mass balances were performed with the carbon material balance closed between 97 and 103 %. This criterion was adopted since compounds containing carbon and hydrogen might accumulate in the reactor in the form of high molecular weight hydrocarbons.

3 Size dependence kinetic model

Based on Parmon–Murzin approach, adsorption on nanoclusters changes the Gibbs free energy [20, 21, 23, 38, 39]. This phenomenon should be accounted for the chemical potential changes:

$$\begin{aligned}\Delta G_{\text{ads}}(r) &= \Delta G_{\text{ads},\infty} \pm \mu_i(r) \mp \mu_{\infty} = \Delta G_{\text{ads},\infty} \pm \delta(r) \\ &= \Delta G_{\text{ads},\infty} \pm \frac{2\sigma V_M}{r}\end{aligned}\quad (1)$$

where, r , μ_{∞} , $\delta(r)$, V_M , and σ are catalyst active site dimension, standard chemical potential of the bulk phase, chemical potential increment, partial molar volume of the substance forming the condensed phase, and surface tension, respectively. Adsorption free energy on nanosize particles compared to that of the bulk material (here, the bulk materials refer to particles with micrometer or larger diameters) is dependent on interactions between adsorbed molecule and surface, and may be changed from positive or negative values.

If K_{ads} increases with decreasing particle size, the adsorption on the nanoparticle surface is stronger than bulk surface. Thus, the adsorption free energy due to nanosize particles is negative, and the change in free energies for these particles can be written as: $\Delta G_{\text{ads}}(r) = \Delta G_{\text{ads},\infty} - \delta(r)$ and thermodynamic adsorption constant (K_{ads}) can be reported as a size-dependent equation:

$$\begin{aligned}\Delta G_{\text{ads}}(r) &= -RT \ln K_{\text{ads}}(r) = \Delta G_{\text{ads},\infty} - \frac{2\sigma V_M}{r} \\ &= -RT \ln K_{\text{ads},\infty} - \frac{2\sigma V_M}{r}\end{aligned}\quad (2)$$

and:

$$K_{\text{ads}}(r) = K_{\text{ads},\infty} \exp\left(\frac{2\sigma V_M}{rRT}\right) = K_{\text{ads},\infty} \exp\left(\frac{\eta}{r}\right)\quad (3)$$

where $K_{\text{ads},\infty}$ is size-independent part of thermodynamic adsorption constant, and η is a parameter which is equal to $\eta = 2\sigma V_M/RT$. The size effect on adsorption kinetics is derived through the Brønsted–Polanyi relation $k = gK^{(1-\chi)}$ and $k_- = gK^{-\chi}$ where k is the rate constant, $K = k/k_-$ is thermodynamic equilibrium constant, g and χ are the

Brønsted–Polanyi parameters, and $0 < \chi < 1$. This relation was used for the explanation of structural activity dependence in homogeneous and heterogeneous catalysis. Thus, the adsorption rate constant for forward and reverse reactions can be reported as [20–24, 40]:

$$\begin{aligned}k(r) &= gK^{(1-\chi)} = gK_{\text{ads},\infty}^{(1-\chi)} \exp\left(\frac{(1-\chi)\eta}{r}\right) \\ &= k_{\infty} \exp\left(\frac{(1-\chi)\eta}{r}\right)\end{aligned}\quad (4)$$

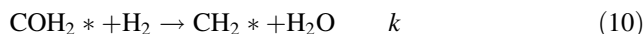
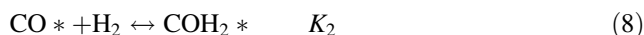
$$k_-(r) = gK^{-\chi} = gK_{\text{ads},\infty}^{-\chi} \exp\left(\frac{-\chi\eta}{r}\right) = k_{-\infty} \exp\left(\frac{-\chi\eta}{r}\right)\quad (5)$$

The kinetics of the Fischer–Tropsch reaction with iron-based catalysts has been studied by many investigators and the development of kinetic models on the basis of a certain mechanism has led to several kinetics models. The observed rate of CO consumption and the rate of synthesis gas consumption ($\text{H}_2 + \text{CO}$) were fitted to the various linearized, kinetic models proposed in literature.

Huff and Satterfield [41] proposed an FTS rate equation with water inhibition on a fused iron catalyst, based on enol/carbide theory as shown below:

$$R_{\text{FTS}} = \frac{kP_{\text{CO}}P_{\text{H}_2}^2}{P_{\text{CO}}P_{\text{H}_2} + \frac{K_3}{K_1K_2}P_{\text{H}_2\text{O}}}\quad (6)$$

This equation is obtained from a set of basic reaction based on combined enol/carbide mechanism, as follows. The rate-determining step is assumed to be the final hydrogenation of the CO-H_2 complex to complete the rupture of the C–O bond.



Based on this FTS rate expression, changes in catalyst FTS activity may be due to changes in: (1) rate constant (k), (2) adsorption parameter (b), and (3) partial pressure of water. The reaction rate expression given in Eq. (1) can be shown as:

$$R_{\text{FTS}} = \frac{kP_{\text{CO}}P_{\text{H}_2}^2}{P_{\text{CO}}P_{\text{H}_2} + bP_{\text{H}_2\text{O}}}\quad (11)$$

where b is adsorption parameter and based on previous set of reaction can be shown as:

$$b = \frac{K_3}{K_1K_2}\quad (12)$$

As discussed in previous paragraph, the Gibbs free energy for steps 1, 2 and 3 in proposed mechanism by Huff

and Satterfield [41] can be written as: $\Delta G_1 = \Delta G_{1\infty} - \delta(r)$, $\Delta G_2 = \Delta G_{2\infty} - \delta(r)$, $\Delta G_3 = \Delta G_{3\infty} - \delta(r)$. Following the same approach as discussed for the size dependence adsorption rates and use of Eqs. (8) and (10), one can write:

$$\frac{k_1(r)}{k_{-1}(r)} = \frac{k_1 \exp((1-\chi)\eta/r)}{k_{-1} \exp(-\chi\eta/r)} = K_1 \exp(\eta/r) \quad (13)$$

$$\frac{k_2(r)}{k_{-2}(r)} = \frac{k_2 \exp((1-\chi)\eta/r)}{k_{-2} \exp(-\chi\eta/r)} = K_2 \exp(\eta/r) \quad (14)$$

$$\frac{k_3(r)}{k_{-3}(r)} = \frac{k_3 \exp((1-\chi)\eta/r)}{k_{-3} \exp(-\chi\eta/r)} = K_3 \exp(\eta/r) \quad (15)$$

Using Eq. (12) for adsorption parameter (b), one can write the size dependence of (b) parameter as:

$$b = \frac{K_3}{K_1 K_2} \exp(\eta/r) = b_\infty \exp(\eta/r) \quad (16)$$

For evaluation of Gibbs free energy of CH_2 insertion (fourth reaction), one can consider the overall reaction as below:



Thus, the overall Gibbs free energy for carbide formation can be evaluated as below:

$$\Delta G_4 = \Delta G_1 + \Delta G_2 + \Delta G_3 \quad (18)$$

$$\Delta G_4 = \Delta G_{4\infty} - 3\delta(r) \quad (19)$$

Thus, rate constant (k) of FTS reaction rates based on previous mechanism can be written as below:

$$k(r) = gK^{3(1-\chi)} = k_\infty \exp\left(\frac{3(1-\chi)\eta}{r}\right) \quad (20)$$

Using Eqs. (16) and (20), one can evaluate the size-dependent thermodynamics parameters by considering the experimental results. The parameters of model were calculated with the Levenberg–Marquardt (LM) algorithm. The mean absolute relative residual (MARR) is reported as a measure of the goodness of the fit:

$$\text{MARR} = 100 \sum_1^n \left| \left(\frac{R_{\text{exp}} - R_{\text{mod}}}{R_{\text{exp}}} \right) \right| \frac{1}{n} \quad (21)$$

where n is the number of data points included.

4 Results and discussion

4.1 Fischer–Tropsch activity

During the activation of iron catalysts, hematite transformed to magnetite (Fe_3O_4) and then to iron carbides. It should be noted that, Fe carbide formation is a necessary step to yield the actual FTS catalyst [14, 29, 42–44]. Thus, iron carbide particle size plays an important role in FTS

reaction and must be considered in size-dependent kinetic evaluations. When hematite is converted to iron carbide, volumetric changes occurred in catalyst particles. Thus, Eqs. (16) and (20) could be used to explain the dependence of the reaction rate on the iron carbides particle size (which was produced after activation.)

The FTS reaction yields organic compounds and the water and/or carbon dioxide as by-products. Thus, carbon monoxide can be consumed for the formation of organic compounds or carbon dioxide. Therefore, the rate of FTS reaction must be written as:

$$r_{\text{FTS}} = r_{\text{CO}} - r_{\text{CO}_2} \quad (22)$$

where r_{CO_2} is the rate of CO_2 formation (water–gas shift reaction) and r_{CO} is the rate of CO consumption. Some experimental results were reported in previous works [45–47]. The observed FTS rates are shown in Fig. 1. In this figure, the effect of temperature and catalyst particle size on FTS reaction rates has been shown and parameter (r) is the radius of active phase. As shown in Fig. 1, the FTS reaction rate increased by increasing the reaction

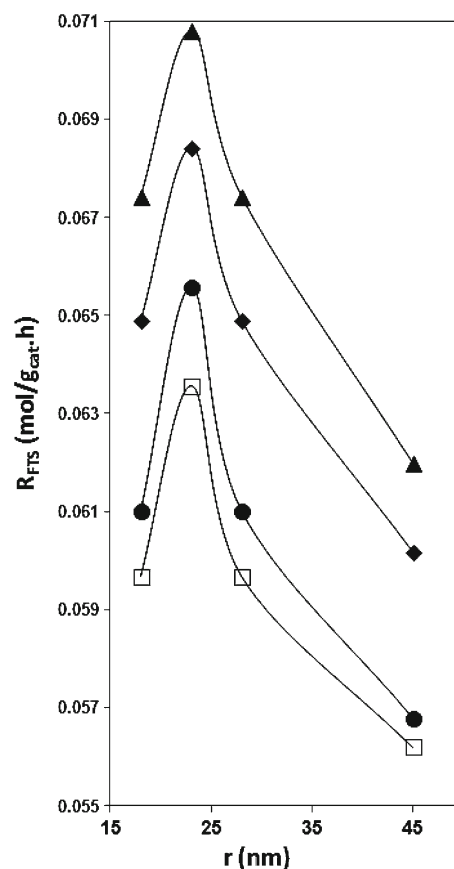


Fig. 1 The effects of temperature and catalyst particle size on FTS reaction rates as function of catalyst particle size (r , nm), (filled triangle $T = 603$ K, filled square $T = 583$ K, filled circle $T = 563$ K, open square $T = 543$ K)

temperature and passed from a maximum by decreasing the catalyst particle size.

The experimental FTS reaction rates fitted by the various linearized kinetic models are proposed in literature. For iron-based catalysts, the equation proposed by Huff and Satterfield [41] [Eq. (4)] is the best kinetic model, especially in high temperatures (about 550 K). The FTS reaction rate expression given in Eq. (6) is linearized by rearrangement as:

$$\frac{P_{H_2}}{-r_{CO+H_2}} = \frac{1}{k} + \frac{b}{k} \frac{P_{H_2O}}{P_{CO}P_{H_2}} \quad (23)$$

Hence, a plot of $\frac{P_{H_2}}{-r_{CO+H_2}}$ versus $\frac{P_{H_2O}}{P_{CO}P_{H_2}}$ should give a straight line with intercept of $(1/k)$ and slope of (b/k) . Figure 2 shows the linearized plots for the model proposed by Huff and Satterfield [41] for the various iron-based catalysts. As shown in Fig. 2, the kinetic data obtained from various iron catalysts in our laboratory are good and represented by the rate equation proposed by Huff and Satterfield [41].

The calculated rate constant (k) and the adsorption parameter (b) at various temperatures are listed in Table 1 and shown in Figs. 3 and 4. These figures and Table 1 showed that by decreasing the catalyst particle size, both the rate constant (k) and the adsorption parameter (b) for FTS reaction are increased simultaneously. These results show the complicated manner of FTS rate as function of the catalyst particle size. The rate equation proposed by Huff and Satterfield predicted that FTS rate over the iron catalyst increased by increasing the FTS rate constant (k) and decreased by increasing the adsorption parameter (b). As shown in Fig. 1, by decreasing the catalyst particle size, the FTS reaction rate passed from a maximum value. Before the maximum point, the FTS reaction is affected by the rate constant (k) and is increased by increasing the rate constant (k). Also, after the maximum point, the FTS reaction is affected by the adsorption parameter (b) and is decreased by increasing the adsorption parameter (b).

The activation energy of FTS reaction is determined from calculated rate constant (k), using the Arrhenius equation model:

Fig. 2 The linearized plots for the model proposed by Huff and Satterfield for the various iron-based catalysts (Pi, bar; RFTS, mol g_{cat}⁻¹ h⁻¹)

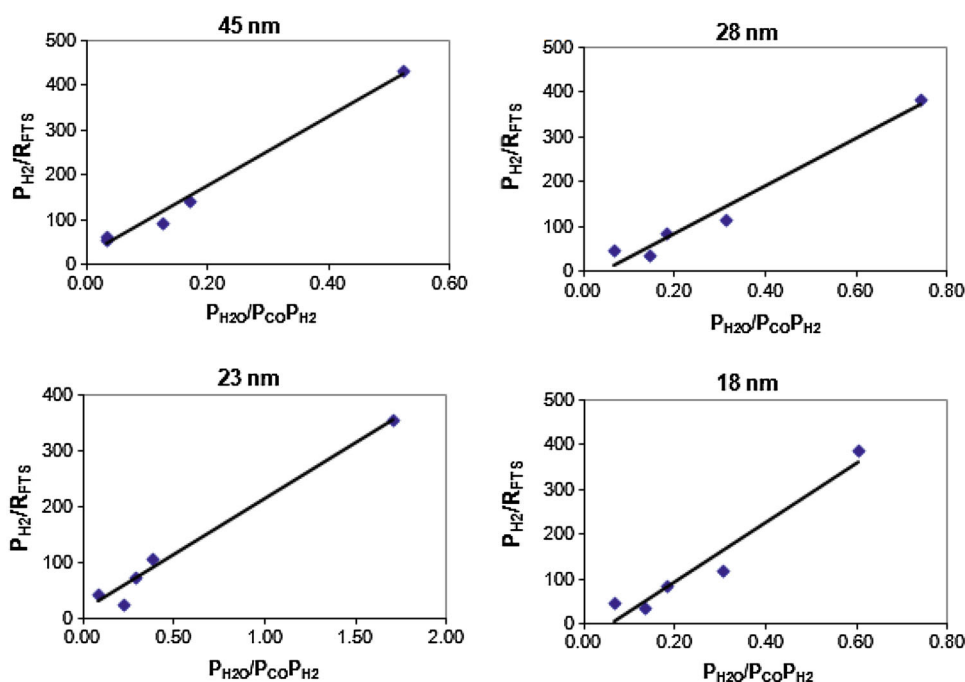


Table 1 Calculated kinetic parameters using Huff and Satterfield FTS kinetic rate equation

Catalyst particle size (nm)	k (mol/gcat h bar)			b (bar)			E_a (kJ/mol)	ΔH (kJ/mol)
	583 K	563 K	543 K	583 K	563 K	543 K		
45	0.829	0.356	0.146	285	205	98	114	71
28	1.020	0.477	0.215	294	211	112	102	64
23	1.241	0.580	0.291	318	235	125	95	62
18	1.360	0.625	0.350	406	285	187	89	51

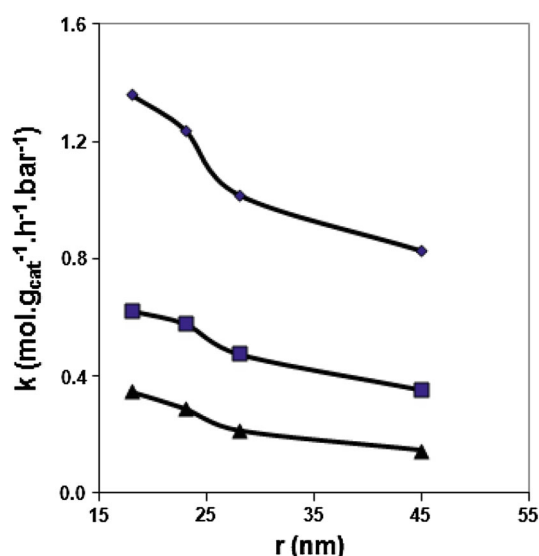


Fig. 3 The calculated FTS rate constant (k , $\text{mol g}_{\text{cat}}^{-1} \text{h}^{-1} \text{bar}^{-1}$) at various temperatures for catalysts as function of catalyst particle size (r , nm), (filledsquare $T = 583$ K, filledrectangle $T = 563$ K, filledtriangle $T = 543$ K)

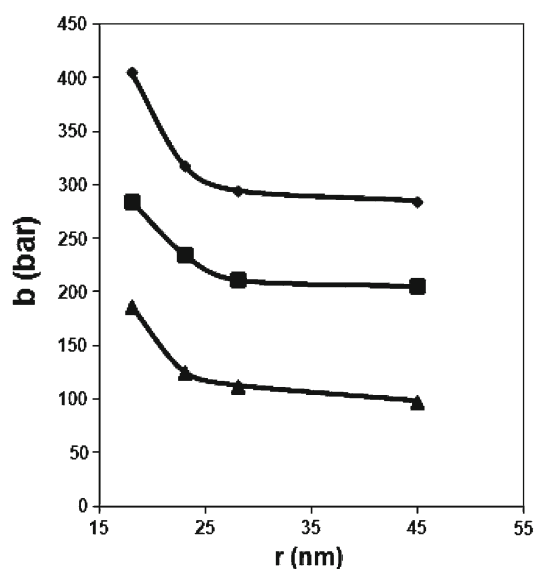


Fig. 4 The calculated adsorption parameter (b , bar) at various temperatures for catalysts as function of catalyst particle size (r , nm), (filledsquare $T = 583$ K, filledrectangle $T = 563$ K, filledtriangle $T = 543$ K)

$$k = k_0 \exp\left(\frac{-E_a}{RT}\right) \quad (24)$$

The calculated activation energy for the FTS reaction is listed in Table 1. As shown in Table 1, the activation energies of catalysts increased from 89 to 114 kJ/mol by increasing the catalyst particle size (r) from 18 to 45 nm. These activation energies are in excellent agreement with

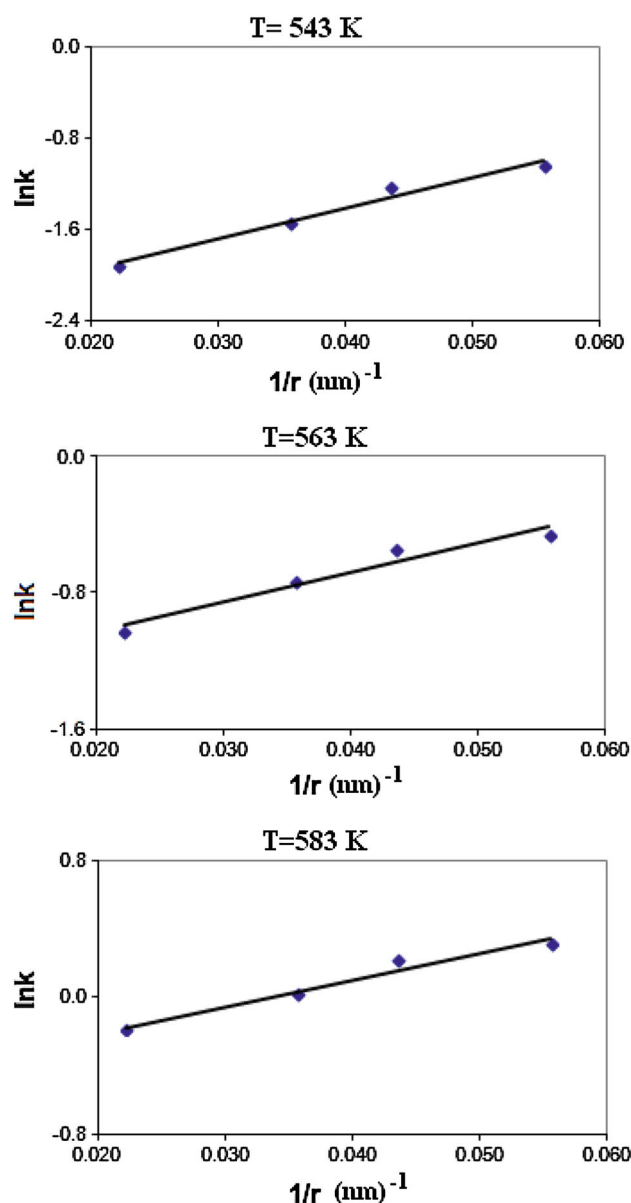


Fig. 5 The linearized plots of the $\ln k$, (k , $\text{mol g}_{\text{cat}}^{-1} \text{h}^{-1} \text{bar}^{-1}$) versus reverse catalyst particle size ($1/r$, nm^{-1}) for the various iron based catalysts

those reported for equivalent values reported in literatures [15, 16, 18, 48, 49].

Adsorption enthalpy ΔH_{ads} is determined with adsorption parameter (b) via another the Arrhenius type equation:

$$b = b_0 \exp\left(\frac{-\Delta H_{\text{ads}}}{RT}\right) \quad (25)$$

The apparent heat of adsorption for the overall catalytic process is listed in Table 1. The apparent heats of adsorption of catalysts calculated about 51 to 71 kJ/mol, are in agreement with those reported for equivalent values reported in literatures [15, 16, 18, 48, 50]. The adsorption

Table 2 Calculated thermodynamic size dependence parameters

b_{cal} (bar)					σ (Jcm ⁻²)	χ	k_{∞} mol g _{cat} ⁻¹ h ⁻¹ bar ⁻¹	η (nm)	b_{∞} (bar)
T	45 nm	28 nm	23 nm	18 nm					
583	269	308	334	394	1,052	0.49	0.596	10.2	214
563	203	238	260	300	1,163	0.50	0.251	11.6	157
543	90	115	133	165	1,754	0.50	0.082	18.2	60

Table 3 Calculated activation energies and enthalpies using size dependence parameters

E_{aCal} (kJ/mol)				ΔH_{Cal} (kJ/mol)				δ_{∞} kJ/cm ²	ΔH_{∞} kJ/mol	$E_{\text{a}\infty}$ kJ/mol
45 nm	28 nm	23 nm	18 nm	45 nm	28 nm	23 nm	18 nm			
112	101	95	86	73	66	62	55	11	85	130

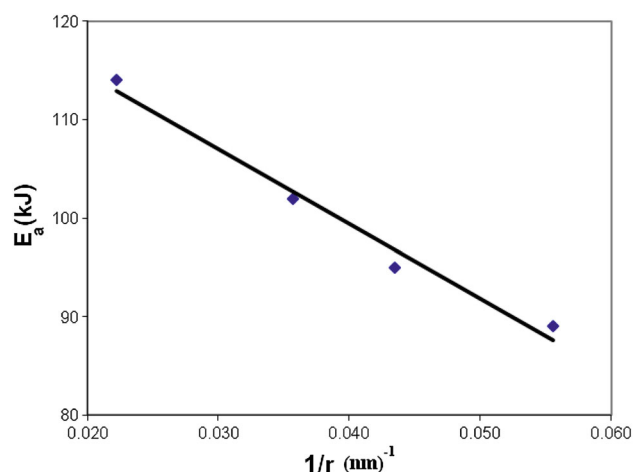
parameter (b) in kinetic model proposed by Huff and Satterfield [41] is equal to $b \approx K_{\text{H}_2\text{O}}/K_{\text{CO}}$ (Eq. 12), where $K_{\text{H}_2\text{O}}$ and K_{CO} are adsorption equilibrium constants for water and carbon monoxide, respectively. Since the apparent heat of adsorption for (b) parameter is 51–71 kJ/mol, the heat of adsorption for water by this model is 51–71 kJ/mol larger than that value for carbon monoxide adsorption [15]. As shown in Table 1, the heat of adsorption increased by increasing the catalyst particle size (r).

4.2 Size dependence evaluation

For evaluation of thermodynamic size dependence parameters, such as surface tension σ and η , the size dependence FTS rate constant given in Eq. (20) is linearized by rearrangement as:

$$\ln k(r) = \ln k_{\infty} + \frac{3(1-\chi)\eta}{r} \quad (26)$$

Hence, a plot of $\ln k(r)$ versus $(1/r)$ should give a straight line with intercept of $\ln k_{\infty}$ and slope of $3(1-\chi)\eta$. Figure 5 shows the linearized plots of the Eq. (26) for the various iron-based catalysts. For the numerical data fitting, the value of Polanyi parameter (χ) was taken to be about 0.5, which is normal value for this parameter [20–22]. The η is a parameter equal to $\eta = 2\sigma V_{\text{M}}/RT$, and the surface tension energy (σ) for iron catalyst can be calculated from experimental data. The calculated thermodynamic size dependence parameters are listed in Table 2. As shown in this table, both size independent FTS reaction rate constant (k_{∞}) and size independent adsorption parameter (b_{∞}) increased by increasing the reaction temperature. Also, Table 2 dictated that the η parameter and surface tension energy (σ) for iron carbides decreased by increasing the reaction temperature.

**Fig. 6** Activation energies for various catalysts as function of reverse catalyst particle size ($1/r$, nm⁻¹)

In Table 2, the adsorption parameter (b_{cal}) is calculated by Eq. (16) using the surface tension energy (σ) for iron carbides [which is derived from experimental results and Eq. (26)]. As shown in Tables 1 and 2, calculated adsorption parameters (b_{cal}) in Table 2 are in good agreement with experimental data in Table 1. These results explained that calculated surface tension energy (σ) for iron carbides using Eq. (26) is correct and can be used for prediction of size-dependent adsorption parameters.

Table 3 listed the size independent activation energy (E_{∞}) and size independent adsorption enthalpy (ΔH_{∞}). These parameters calculated from two ways, first extrapolated the plot of size-dependent activation energies (Fig. 6) and size-dependent adsorption enthalpies (Fig. 7) against $(1/r)$ to 0 ($r = \infty$). In second way, the size independent FTS reaction rate constant (k_{∞}) and size independent adsorption parameter (b_{∞}) were calculated from the Arrhenius equation. Figure 8 shows the plot of

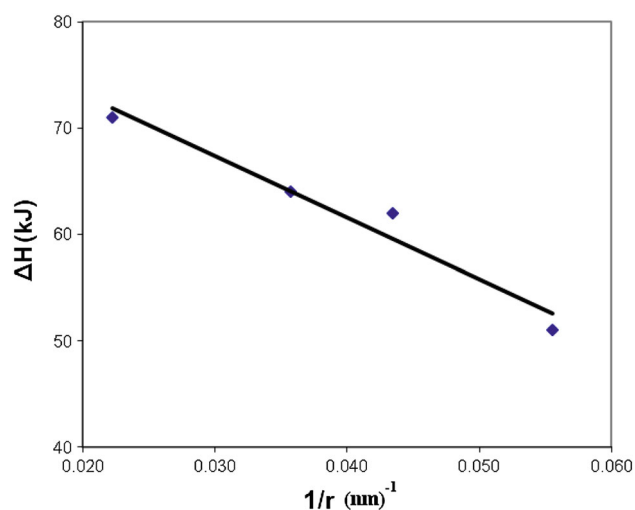


Fig. 7 Adsorption enthalpy ΔH_{ads} as function of reverse catalyst particle size ($1/r$, nm^{-1})

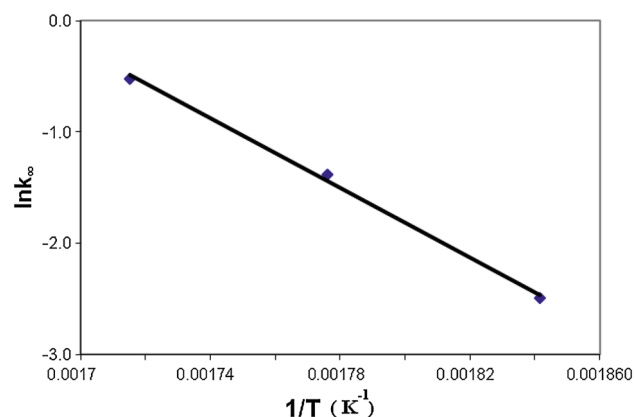


Fig. 8 Size independent FTS reaction rate constant (k_{∞}) (k , $\text{mol g}_{\text{cat}}^{-1} \text{ h}^{-1} \text{ bar}^{-1}$) as function of reverse temperature ($1/T$, K^{-1})

$\ln(k_{\infty})$ versus ($1/T$) that gives a straight line with slope of $(-E_{\infty}/R)$ and Fig. 9 shows the plot of $\ln(b_{\infty})$ versus ($1/T$) that gives a straight line with slope of $(-\Delta H_{\infty}/R)$. As listed in Table 3, both the methods give the same size independent activation energy (E_{∞}) and size independent adsorption enthalpy (ΔH_{∞}), which are equal to 130 and 85 kJ/mol, respectively.

For evaluation of size-dependent activation energy and adsorption enthalpy of iron catalysts, the absolute temperature independent surface tension energies (σ_{∞}) must be calculated. From temperature-dependent results of η ($\eta = 2\sigma V_M/RT$) which is listed in Table 3, the temperature independent surface tension energy (σ_{∞}) for iron catalyst can be calculated. Figure 10 showed the variation of η against the temperature, and the surface tension energy (σ_{∞}) for iron catalyst is calculated from the slope of this plot. The surface tension energy for iron catalyst is

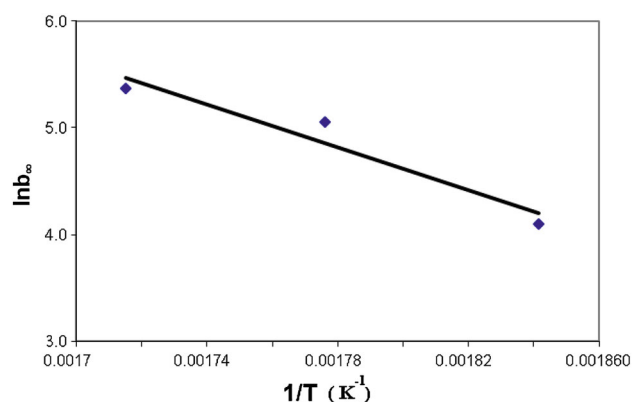


Fig. 9 Size independent adsorption parameter (b_{∞} , bar) as function of reverse temperature ($1/T$, K^{-1})

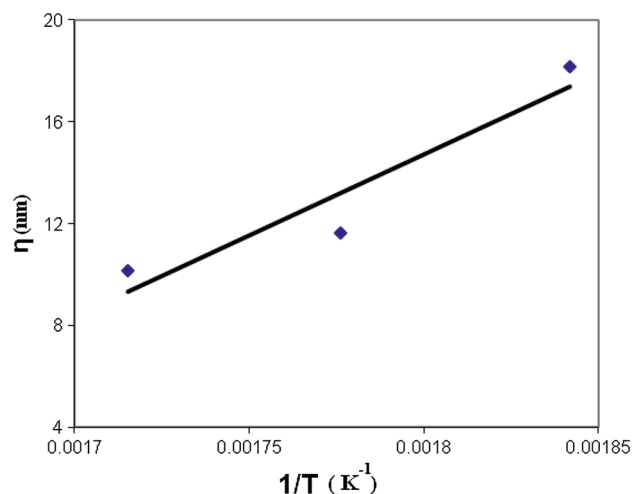


Fig. 10 variation of η (nm) as a function of reverse temperature ($1/T$, K^{-1})

obtained about 11 kJ/cm², which is the same as previous literature reports [20–22]. Equations (16) and (20) predicted that the size dependence activation energy and adsorption enthalpy of iron catalysts can be evaluated as below:

$$\Delta H(r) = \Delta H_{\infty} - \frac{2\sigma V_M}{r} \quad (27)$$

$$E_a(r) = E_{a\infty} - \frac{6(1-\chi)\sigma V_M}{r} \quad (28)$$

The results for size-dependent activation energy and adsorption enthalpy of iron catalysts are listed in Table 3. By comparison with experimental results in Table 1, it is seen that the calculated activation energy and adsorption enthalpy are in agreement with experimental results, for various catalysts with different particle size.

5 Conclusion

Thermodynamic analysis of the particle size effects on kinetics parameters of FTS reaction is studied on lanthanum promoted iron catalysts. The experimental results showed the FTS reaction rates passed from a maximum value by decreasing the catalyst particle size. By decreasing the catalyst particle size, not only adsorption equilibrium but also kinetic rate constant increased. The results showed that the activation energies of FTS reaction increased from 89 to 114 kJ/mol and heats of adsorption of catalysts increased from 51 to 71 kJ/mol by increasing the iron particle from 18 to 45 nm. The key point in thermodynamic analysis is description of influence of the nanometric size of particles on the chemical potential of the active phase (e.g., clusters supported on a carrier).

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